LOKMANYA TILAK JANKALYAN SHIKSHAN SANSTHA'S

PRIYADARSHINI COLLEGE OF ENGINEERING

(An Autonomous Institution Affiliated to R.T.M. Nagpur University)

Department of Information Technology

Semester :- VII (2026-2027)

Title :- An AI-Powered Personalized Pregnancy Wellness & Family Support Platform

Nurture AI

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Introduction

Pregnancy is a crucial phase that requires regular health monitoring, proper nutrition, and timely medical guidance.

Nurtur AI is an AI-powered pregnancy wellness platform that provides personalized health recommendations, AI-based symptom analysis, nutrition guidance, smart reminders, digital health records, and family support. By integrating these features into a single platform, Nurtur AI aims to make pregnancy care smarter, more personalized, and more accessible.

Research Objective

Our aim is to build a platform that improves pregnancy care by using AI, keeping all health information in one place, and providing personalized support to both the mother and her family.

Research Gap

- Generic AI recommendations
- Limited health data analysis
- Basic pregnancy risk prediction
- Non-personalized nutrition guidance
- Limited family participation
- Scattered pregnancy records
- Reactive instead of proactive healthcare
- Lack of an integrated AI-powered pregnancy platform

Proposed Solution

Nurtur AI solves existing problems by integrating multiple intelligent features into one platform:-

🤖 **AI Pregnancy Assistant:-** Provides personalized answers and health guidance.

📊 **Daily Health Score:-** Calculates daily health condition using user inputs.

🥗 **Nutrition Analyzer:-** Tracks food intake and suggests nutritional improvements.

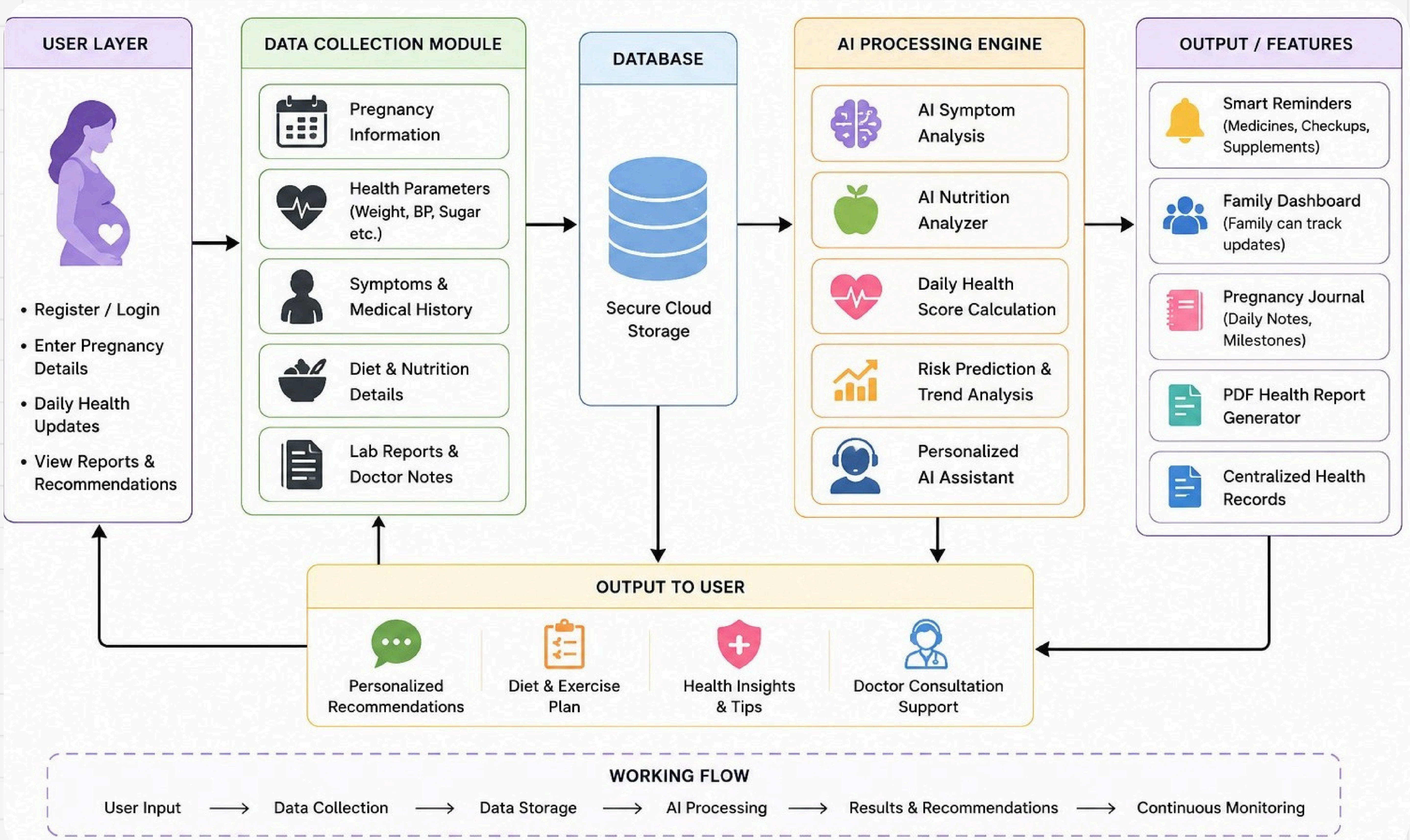
📈 **Symptom Trend Analysis:-** Analyzes symptom patterns to identify risks.

👨‍👩‍👧 **Family Dashboard:-** Allows family members to monitor updates and provide support.

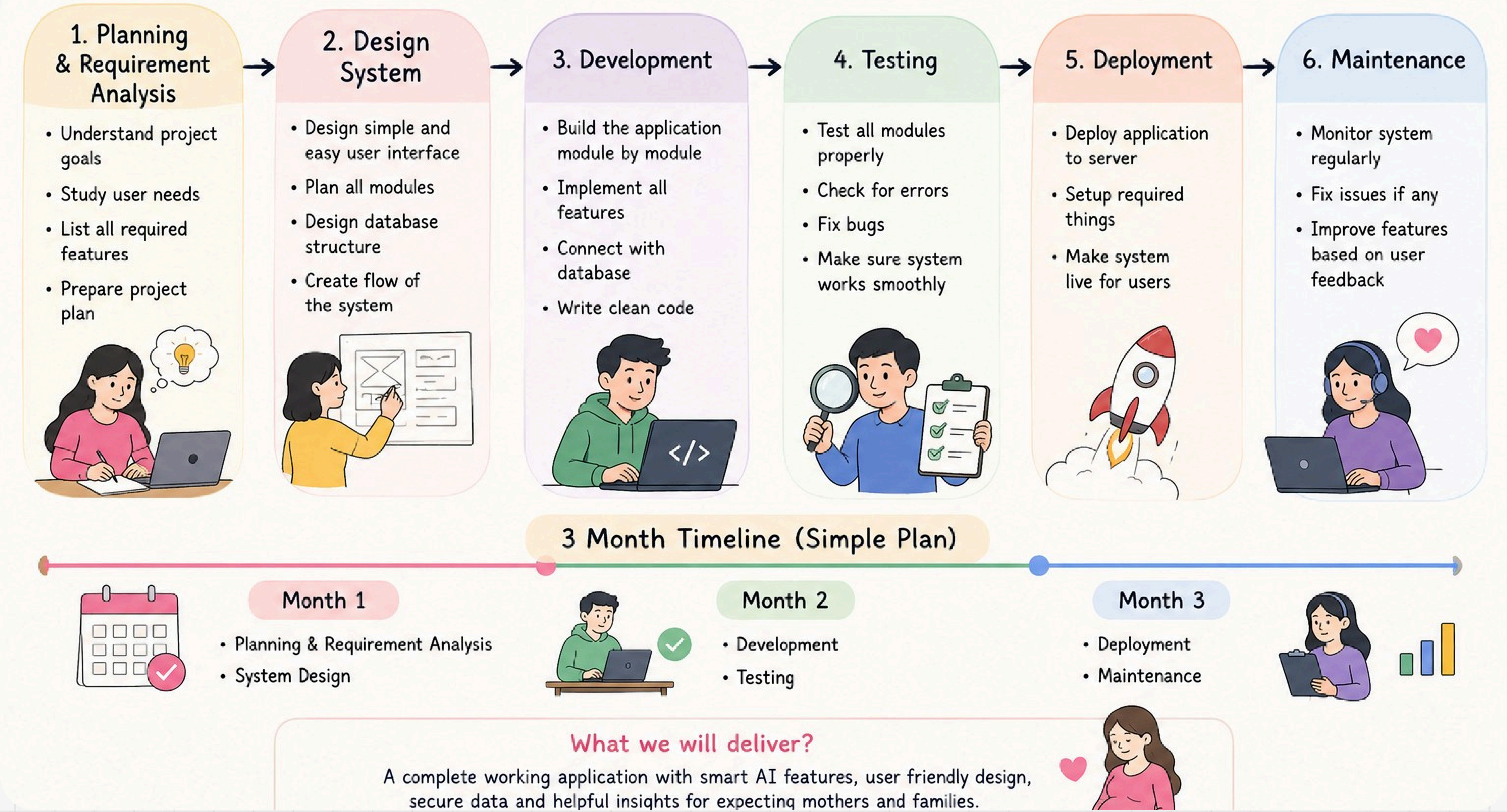
📂 **Digital Pregnancy Records:-** Stores reports and medical history securely.

🔔 **Smart Reminders:-** Sends reminders for medicines, checkups, and water intake.

Flow of Proposed Methodology



Research Plan



Technologies

🖥️ **Frontend:-** React Native, JavaScript, TypeScript
⚙️ **Backend:-** Java, Spring Boot, Spring Data JPA
🤖 **Artificial Intelligence:-** Google Gemini API, Prompt Engineering
🗄️ **Database:-** MySQL
☁️ **Cloud Storage:-** Firebase Storage / AWS S3
🔔 **Notifications:-** Firebase Cloud Messaging
🧪 **API Testing:-** Postman
💻 **Development Tools:-** IntelliJ IDEA, Android Studio, Visual Studio Code
🔄 **Version Control:-** Git, GitHub
📦 **Build Tool:-** Maven
🚀 **Deployment:-** Render/Railway, MySQL Server, Google Play Store

References

1. Daley, B. J., et al. (2022). mHealth Apps for Gestational Diabetes Mellitus that Provide Clinical Decision Support or Artificial Intelligence: A Scoping Review. Diabetic Medicine.

2. Wei, H. X., Yang, Y. L., Luo, T. Y., & Chen, W. Q. (2023). Effectiveness of Mobile Health Interventions for Pregnant Women with Gestational Diabetes Mellitus: A Systematic Review and Meta-analysis. Journal of Obstetrics and Gynaecology.

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Thank You